

A PROJECT REPORT ON

Diet Recommendation System

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ABSTRACT

*The **Diet Recommendation System** is designed to provide personalized dietary guidance to individuals seeking to improve their health and nutrition. In light of the increasing prevalence of lifestyle-related diseases such as obesity and diabetes, this project aims to address the challenges associated with maintaining a healthy diet. By integrating user-specific health data, preferences, and nutritional requirements, the system generates customized meal plans that cater to each individual's unique needs.*

Utilizing advanced machine learning algorithms, the system analyzes extensive nutritional databases and user inputs to deliver accurate and relevant recommendations. A key feature is the incorporation of a feedback mechanism, allowing the system to adapt dynamically to changing health metrics and user preferences over time. This holistic approach not only enhances user engagement but also promotes sustainable health behavior changes, empowering individuals to make informed dietary choices. Ultimately, the Diet Recommendation System aims to foster healthier lifestyles and improve overall well-being.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The Diet Recommendation System is designed to provide personalized dietary guidance to individuals seeking to improve their health and nutrition. In light of the increasing prevalence of lifestyle-related diseases such as obesity and diabetes, this project aims to address the challenges associated with maintaining a healthy diet. By integrating user-specific health data, preferences, and nutritional requirements, the system generates customized meal plans that cater to each individual's unique needs. Utilizing advanced machine learning algorithms, the system analyzes extensive nutritional databases and user inputs to deliver accurate and relevant recommendations. A key feature is the incorporation of a feedback mechanism, allowing the system to adapt dynamically to changing health metrics and user preferences over time. This holistic approach not only enhances user engagement but also promotes sustainable health behavior changes, empowering individuals to make informed dietary choices. Ultimately, the Diet Recommendation System aims to foster healthier lifestyles and improve overall well-being.

1.2 Motivation

The motivation behind the Diet Recommendation System stems from the rising incidence of lifestyle-related diseases, such as obesity and diabetes, which underscore the necessity of maintaining a healthy diet. Many individuals struggle with traditional dietary guidelines due to a lack of personalization and adaptability, making it challenging to adhere to nutritional recommendations. By harnessing the power of machine learning, this project aims to offer tailored dietary solutions that consider individual health metrics, preferences, and goals. The system seeks to bridge the gap between general dietary advice and specific user needs, thereby enhancing engagement and compliance. Additionally, the incorporation of real-time feedback ensures that recommendations remain relevant and effective over time. Ultimately, the goal is to empower individuals to take charge of their nutritional choices, leading to healthier eating habits and improved overall health outcomes. This personalized approach not only promotes individual well-being but also contributes to the broader public health landscape.

1.3 Problem Definition and Objectives

The increasing prevalence of lifestyle-related diseases, such as obesity, diabetes, and cardiovascular conditions, emphasizes the need for a healthy diet. However, many individuals struggle to follow generic dietary guidelines that fail to account for personal preferences, cultural backgrounds, and specific health conditions. Traditional dietary recommendations often lack the necessary personalization, making it challenging for users to adopt and maintain healthy eating habits. Additionally, existing dietary recommendation systems typically rely on fixed rules and historical data, rendering them inflexible and less effective in addressing users' dynamic needs. This can lead to frustration and disengagement from healthy eating practices, further exacerbating health issues. Moreover, there is a significant gap in integrating real-time health data from sources like wearable devices and personal health records, limiting the ability of dietary systems to provide timely and relevant suggestions. Users also face challenges in trusting these systems due to a lack of transparency in how recommendations are generated. The Diet Recommendation System aims to address these challenges by offering personalized, adaptable, and transparent dietary suggestions. By integrating user-specific data and leveraging machine learning algorithms, this system empowers individuals to make informed dietary choices, promoting better health and well-being.

- To create a system that gives diet suggestions based on a person's age, weight, health conditions, and food likes.
- To use machine learning methods to make diet plans more accurate and personalized.
- To update diet plans automatically when a user's health or preferences change.
- To help users achieve goals like weight loss, better fitness, or managing diseases through better eating.
- To improve recommendations by learning from user feedback over time.

1.4 Project Scope & Limitations

Project Scope : Personalized Diet Plans: The system will generate customized meal plans based on user inputs such as age, gender, weight, health

conditions, activity levels, dietary preferences (e.g., vegetarian, vegan, gluten-free), and fitness goals (e.g., weight loss, muscle gain, or diabetes management). **Health Data Integration:** Users can integrate their health metrics from wearable devices (e.g., Fitbit, Apple Watch) or manually input data such as weight, blood pressure, and glucose levels to get more accurate and real-time dietary recommendations. **Machine Learning Algorithms:** Machine learning models will analyze user data, predict dietary needs, and optimize meal recommendations based on past user preferences and health patterns, ensuring continuous improvement in the accuracy and relevance of the meal plans. **Nutritional Information:** The system will provide detailed nutritional breakdowns of recommended meal plans, including calories, macronutrients (protein, carbohydrates, fats), and micronutrients (vitamins, minerals). This helps users make informed decisions about their diet.

Project Limitations : While the Diet Recommendation System aims to offer a highly effective and personalized dietary experience, it does face several limitations: **Dependence on User Input:** The accuracy of the recommendations depends heavily on the completeness and accuracy of user-provided data, such as weight, health conditions, or preferences. Incorrect or incomplete inputs may lead to suboptimal recommendations. **Lack of Real-Time Medical Integration:** Although the system may incorporate data from wearables, it is not a replacement for real-time medical advice from healthcare professionals. It cannot monitor sudden changes in medical conditions, such as acute diseases, and may not provide immediate responses to medical emergencies. **Limited Scope of Health Conditions:** While the system can generate meal plans based on common health goals (e.g., weight loss, diabetes management), it may not cover all possible medical conditions or dietary needs (e.g., rare allergies or complex medical conditions requiring specific dietary interventions). **Generalized Recommendations:** Despite being personalized, the recommendations are based on general nutritional principles and widely accepted dietary guidelines. The system does not take into account highly individualized biochemical responses to food, which vary significantly among people

1.5 Methodologies of Problem solving

The Diet Recommendation System uses a systematic approach to solve the problem of providing personalized dietary recommendations. Below is a detailed explanation of the methodology adopted in this project:

1. **Problem Understanding :** The project begins with understanding the key challenges associated with maintaining a healthy diet, particularly in

the context of varying individual health conditions, dietary preferences, and lifestyle factors. Identifying the main problem: Lack of personalized diet recommendations that consider specific user health metrics, goals, and preferences. Researching existing solutions and limitations, such as generic meal plans and manual diet tracking tools that fail to offer dynamic, personalized guidance.

2. Data Collection and Preprocessing : User Data: Collection of user-specific information like age, gender, weight, height, dietary restrictions, activity level, and health goals (e.g., weight loss, diabetes management). The data can be collected via forms or integrated from wearable devices and health applications.

3. Nutritional Database: Integration of nutritional data for a wide range of food items, including macronutrients, micronutrients, calories, and allergens. This database must be accurate and up-to-date. Preprocessing: Cleaning and formatting user data to ensure it is consistent and usable for further analysis. Missing values are handled, and user inputs are standardized to fit the requirements of machine learning algorithms. Feature Engineering : Extracting and defining relevant features from the collected data, such as Body Mass Index (BMI), daily caloric needs, and nutritional deficiencies. These features help in building an accurate user profile and predicting dietary needs.

4. Algorithm Selection : Machine Learning Models: Selection of appropriate machine learning models to predict dietary needs based on user data. Algorithms such as Decision Trees, Random Forest, or Support Vector Machines (SVM) are used to generate personalized meal plans. These models analyze user inputs in conjunction with the nutritional database to suggest balanced and healthy meals. Optimization Algorithms: Use of optimization techniques (e.g., genetic algorithms) to ensure the meal plans meet specific nutritional goals while considering user preferences and restrictions.

5. Meal Plan Generation : Based on the processed data and algorithms, the system generates meal plans that are tailored to the user's health profile. The generated meal plans will include appropriate portions, nutritional breakdown, and suggested foods. The system ensures that meal plans are nutritionally adequate, culturally relevant, and cost-effective.

CHAPTER 2

LITERATURE SURVEY

1. **Personalized Diet Recommendation System Using ML** (Navya Narayana Kumari et al., 2024)
Uses machine learning models to offer personalized diet suggestions based on user profile and nutritional needs. Focuses on prediction accuracy and real-time recommendations.
2. **Diet Recommendation System Using Machine Learning** (Bhagat & Tuteja, 2023)
Implements Random Forest and Decision Tree algorithms to provide tailored meal plans, considering user-specific health metrics and dietary goals.
3. **Diet Recommendation System Based on User Activities** (Jadhav et al., 2022)
A lifecycle case study for building a recommendation system that integrates user activity data for improving the relevance of diet suggestions.
4. **Expert System for Diet Recommendation** (Kovácsnai, 2011)
Develops a rule-based expert system using ripple-down rules and genetic algorithms to incrementally optimize personalized diet plans.
5. **Systematic Review of Nutrition Recommendation Systems** (Abhari et al., 2019)
Surveys 25 nutrition recommender systems and concludes hybrid models outperform rule-based or content-based approaches in accuracy and personalization.
6. **Health-Aware Food Recommender System** (Ge et al., 2015)
Introduces a food recommender system combining collaborative filtering with health-aware personalization to accommodate diverse dietary needs.
7. **Yum-Me: Nutrient-Based Meal Recommender** (Yang et al., 2017)
Provides personalized meal planning based on individual nutrient requirements and user preferences using a nutrient optimization model.
8. **Intelligent Food Planning: Recipe Recommendation** (Freyne & Berkovsky, 2010)
Uses collaborative filtering to personalize recipe suggestions by learning user preferences and dietary goals.
9. **Towards Health-Aware Recommender Systems** (Elsweiler et al., 2015)

Highlights the importance of health in recommender systems and presents strategies to balance taste preferences and nutritional goals.

10. **Nutritional Recommender Systems: State-of-the-Art** (El-Din & Goher, 2020)

Reviews existing nutritional recommender systems and identifies research gaps and future opportunities in system design and personalization.

11. **Food Recommender Systems: Key Challenges** (Trattner & El-sweiler, 2017)

Outlines the main contributions and future research directions in food recommender systems with a focus on nutrition and health.

Reference Title	Algorithms	Data Sources	Advantages	Limitations
ChatDiet [6]	BM25, Casual Inference, GPT-3.	User input, casual graphs	Lightweight and scales well for large datasets.	Focuses on exact or near-exact matches, which may not capture broader semantics.
Systematic Review of NRS[5]	Hybrid Recommender Systems, Rule-based	NRS datasets, user preference health data	Combines multiple techniques (e.g., collaborative and content-based filtering) to improve recommendation quality.	More complex to implement and tune compared to standalone approaches
Expert System for Diet Recommendation [4]	Ripple Down Rules, CBR, Genetic Algorithms	Patient data, health conditions	Incrementally builds rules, allowing for easy updates and maintenance.	Performance heavily depends on the quality and coverage of the rules.
Systematic Review of Food Recommender Systems [10]	Content-based filtering, Hybrid methods	Food datasets, user preference	Independent of other users' data, ensuring unique recommendations.	Lacks diversity, as it tends to recommend items similar to those already interacted with.
Personalized Diet Recommendation System[1]	Random Forest, K-means Clustering.	A dataset that includes user demographic and health metrics	Capable of processing large datasets	Prone to overfitting without tuning.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

- Users will have access to the internet and are expected to provide accurate health information (e.g., weight, height, dietary preferences).
- The system assumes the availability of cloud infrastructure to store user data and manage meal plans.
- The system relies on a nutritional database and API for up-to-date food composition and dietary guidelines.
- Dependencies include secure authentication systems and APIs for data processing and external services.

3.2 Functional Requirements

3.2.1 User Registration and Authentication Users will register on the platform by providing essential personal information, including name, email, password, age, gender, height, weight, and any dietary restrictions. After authentication via email, users can access the platform.

3.2.2 Meal Plan Creation and Storage Based on user inputs, the system will generate personalized meal plans. These meal plans will be stored in the user's profile and can be modified or updated according to changing user preferences or health goals.

3.2.3 Decentralized File Storage The system will securely store meal plans and user health data in the cloud. For scalability, decentralized storage options could be explored to maintain efficiency and security.

3.2.4 Health Data and Progress Tracking Users can input regular health updates, such as weight changes, and track their progress over time. The system will use this information to adjust meal plans as necessary to meet the user's health goals.

3.2.5 Smart Contract for Meal Plan Management A potential future enhancement involves integrating smart contracts for managing user subscriptions and automated meal plan renewals based on user progress.

3.2.6 Document Retrieval Users should have quick access to stored meal plans, nutritional details, and progress reports. They can retrieve these documents for future reference or share them with healthcare professionals.

3.2.1 System Feature 1(Functional Requirement)

The system shall allow users to create and manage profiles with personal, health, and preference data. Users can update this data anytime, and the system will immediately reflect these changes in future recommendations.

3.2.2 System Feature2 (Functional Requirement)

The system shall use the collected user data to dynamically generate nutritionally balanced and goal-specific meal plans using machine learning models such as Random Forest or SVM.

3.2.3 System Feature n (Functional Requirement)

The system shall incorporate a feedback mechanism where users can rate or modify their recommended meal plans, allowing the system to improve its accuracy over time based on learning from user preferences.

3.3 External Interface Requirements

3.3.1 User Interface The system will feature a user-friendly and intuitive web interface where users can navigate easily between sections like meal plans, progress tracking, and account settings. The interface will support both desktop and mobile versions.

3.3.2 Nutritional Database API Integration The system will interface with an external nutritional database API to fetch data on various foods and their nutritional content. This data will be used to generate accurate meal recommendations.

3.3.3 Cloud Storage Integration The system will leverage cloud-based storage solutions to manage and store user health data, meal plans, and other associated documents.

3.3.1 User Interfaces

The UI will include dashboards for meal suggestions, progress tracking, and account settings. It will be developed using responsive design principles and tested across browsers and devices.

3.3.2 Hardware Requirements

- Client Device: computer with internet access and a modern browser.
- Server: Cloud-based or on-premise server with multi-core CPU, 16GB+ RAM.

3.3.3 Software Requirements

1. Operating System: Anyone among MAC OSX, Linux or Windows
2. Jupyter Notebook
3. Dataset pertinent to medical images
4. Programming Language: Python

3.3.4 Communication Interfaces

The system will use HTTPS for all communications between frontend, backend, and third-party APIs. SMTP protocol will be used for sending notifications. RESTful APIs will handle client-server communication.

3.4 Nonfunctional Requirements

The system must generate meal plans within a few seconds of user input, providing a smooth and responsive experience. It should be able to handle multiple user requests simultaneously and be scalable to accommodate an expanding user base.

3.4.2 Safety and Security Requirements

User data, particularly sensitive health information, must be encrypted during transmission and storage. The system will ensure data privacy compliance with regulations such as GDPR, and access to user data will be controlled through role-based authentication and authorization mechanisms.

3.4.1 Performance Requirements

The system must process and return meal plans in under 3 seconds on average. It should support at least 100 concurrent users with consistent performance under load.

3.4.2 Safety Requirements

The system should detect and warn about contradictory user inputs (e.g., allergy to a recommended food). Automatic data backups and secure storage must be enforced to prevent data loss or corruption.

3.4.3 Security Requirements

All user data must be stored securely using AES encryption and accessed only via authenticated sessions. Use of secure login methods (e.g., JWTs or OAuth 2.0) and periodic vulnerability scans is mandatory.

3.4.4 Software Quality Attributes

The system must be maintainable, scalable, and portable. It should have modular code architecture, detailed documentation, and be easy to deploy on cloud platforms. It must meet usability standards to ensure accessibility and user satisfaction.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The architecture of the Diet Recommendation System is designed to ensure scalability, personalization, and efficient dietary planning using content-based filtering and machine learning algorithms. Users interact with the system through a user-friendly interface developed using Streamlit. They can register, log in, and input personal health data such as age, gender, height, weight, activity level, and dietary preferences.

Upon submission, user data is processed on the backend powered by FastAPI. The system invokes a trained Nearest Neighbors model using cosine similarity to match users with similar dietary patterns and recommend optimal recipes. The recipes and metadata are sourced from a Food.com dataset containing over 500,000 entries.

Nutritional and recipe data is stored in a structured format using CSV files or optionally in a relational database. A unique identifier is associated with each recipe to track selections, feedback, and ratings. The feedback loop further refines future recommendations using updated input.

This modular architecture promotes secure data handling, easy maintenance, and the ability to scale with additional datasets or advanced models.

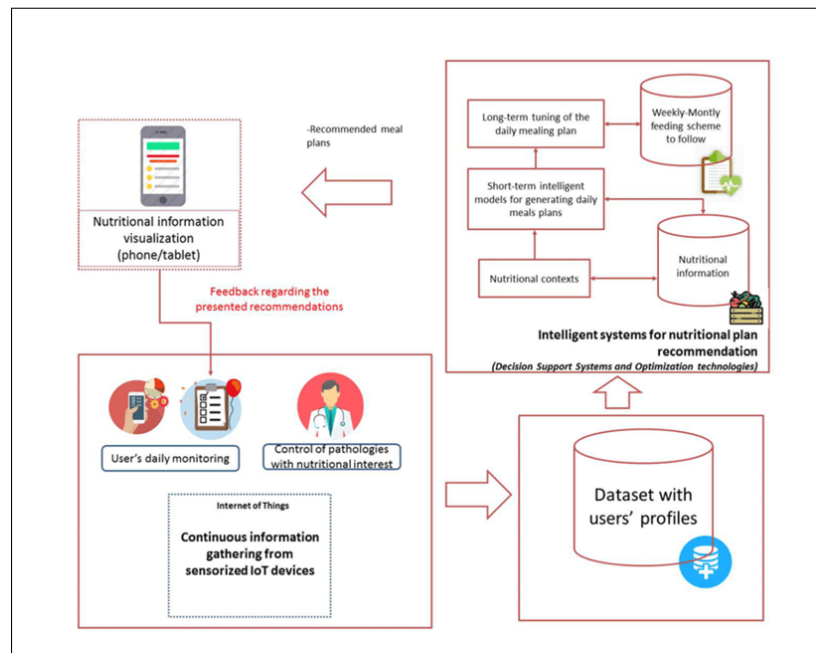


Figure 4.1: System Architecture

4.2 Data Flow Diagrams

- **User Interface (Frontend)** – Users access the platform via a Streamlit-based web interface. – User actions include registration/login, health data entry, viewing recommended meals, and submitting feedback.
- **Authentication** – Users authenticate using a secure login system. Session data is managed by backend logic in FastAPI.
- **Data Input and Processing** – Health metrics and dietary preferences are submitted to the backend. – Data is preprocessed and passed to the ML engine for recommendation generation.
- **ML Recommendation Engine** – Applies cosine similarity using Nearest Neighbors to find best-suited meals. – Returns top recommended recipes based on user profile.
- **Dataset Integration (Food.com)** – Recipes are fetched from a structured dataset with nutritional information. – Each recipe has a unique ID for tracking.
- **Storage and Feedback Loop** – Stores user preferences, ratings, and past selections. – Feedback is used to improve future meal suggestions.

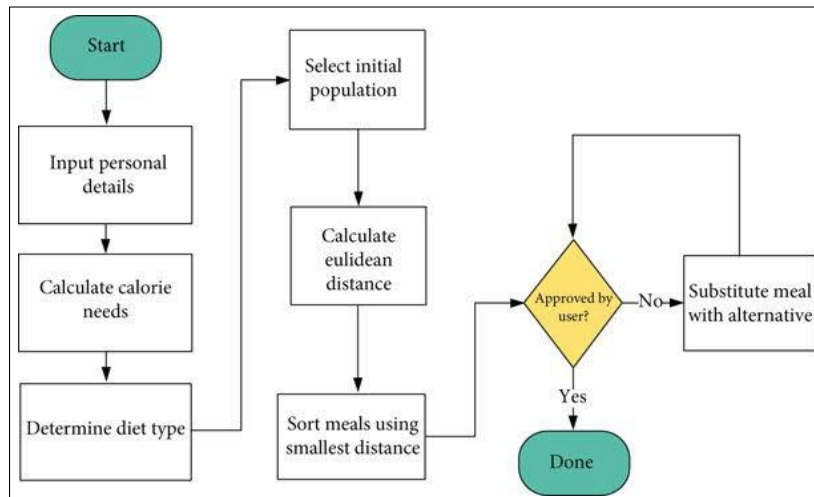


Figure 4.2: Data Flow Diagram

4.3 UML Diagram

The UML diagram in Figure 4.3 illustrates the activity flow for generating and approving a personalized diet plan using a content-based recommendation approach. The process is broken down as follows:

- **Start:** The process begins.
- **Input Personal Details:** The user provides personal data such as age, weight, height, gender, activity level, and dietary preferences.
- **Calculate Calorie Needs:** Based on the input, the system calculates the user's daily calorie requirements using standard nutritional formulas (e.g., Harris-Benedict Equation).
- **Determine Diet Type:** Based on user preferences and calorie needs, the system categorizes the user into a suitable diet type (e.g., vegetarian, high protein, low carb).
- **Select Initial Population:** The system selects an initial pool of meals/recipes from the dataset (e.g., Food.com dataset).
- **Calculate Euclidean Distance:** The system compares the nutritional profile of the meals with the user's profile using Euclidean distance to determine similarity.
- **Sort Meals Using Smallest Distance:** Meals are ranked based on proximity to the user's ideal nutritional requirements — the smaller the distance, the better the match.
- **Approved by User?:** The system presents the top meal options to the user for approval.
 - **Yes:** If the user approves the meal, the process ends.
 - **No:** If the user rejects the meal, the system substitutes it with an alternative (next best option) and loops back for approval.
- **Done:** The process completes when the user approves a meal.

This activity diagram models a decision-based, iterative recommendation loop that ensures personalized and satisfactory meal suggestions based on dietary goals.

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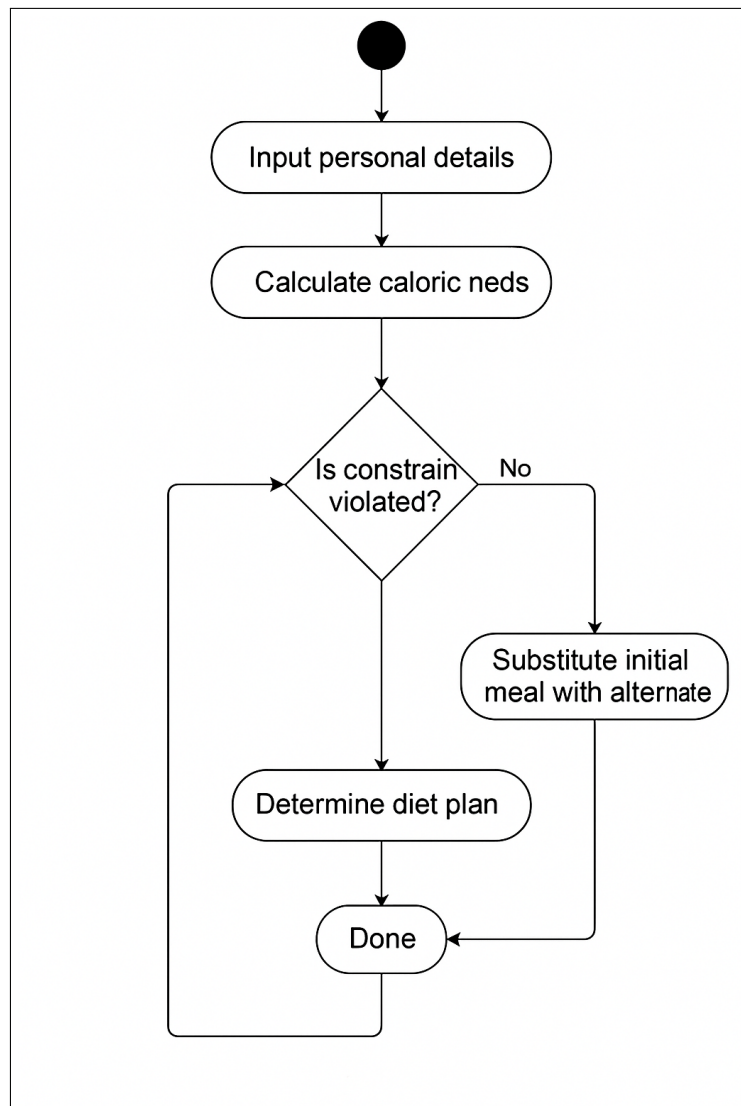


Figure 4.3: UML Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

The project estimate is prepared after considering initial rough estimates and refining them through multiple evaluation cycles (analogous to the reconciliation process). The following aspects were considered for the Diet Recommendation System project:

- **Development Time Estimate:** The project phases were initially estimated based on standard SDLC practices. After prototype development and feasibility analysis, time allocations were revised. Major time blocks were assigned for requirement analysis (10%), data collection and preparation (20%), model development and training (25%), back-end and frontend development (25%), integration and testing (15%), and final deployment (5%).

5.1.2 Project Resources

The successful implementation of the Diet Recommendation System required a combination of human, software, and hardware resources:

- **Project Developer:** Responsible for end-to-end development including model creation, API development, and UI integration.
- **Machine Learning Engineer:** Built the recommendation model using Nearest Neighbors with cosine similarity.

Software Resources:

- **Python (v3.10.8):** Main programming language used across the project.
- **FastAPI (v0.88.0):** Lightweight backend framework to expose APIs for diet recommendations.
- **Uvicorn (v0.20.0):** ASGI server to run the FastAPI backend.
- **Scikit-learn (v1.1.3):** Used to implement the Nearest Neighbors recommendation algorithm.
- **Pandas (v1.5.1) and NumPy (v1.21.5):** For data preprocessing and manipulation.

- **Streamlit (v1.16.0)**: Framework for building an interactive front-end interface.
- **streamlit-echarts (v1.24.1)**: Used to enhance frontend visualizations.
- **BeautifulSoup4 (v4.11.1)**: For any required HTML parsing (if applicable).
- **Docker and Docker-Compose**: Used for containerizing and deploying the frontend and backend as separate services.
- **Food.com Dataset from Kaggle**: Contains over 500,000 recipes and 1.4 million reviews.

Hardware Resources:

- **Development Machine**: Local system with sufficient CPU/RAM to run model training and testing.
- **Docker Runtime Environment**: Used to ensure consistency between development and deployment environments.

The use of these resources helped ensure that the system was easy to build, scalable, and reliable across different environments.

5.2 Risk Management

- Risk management involves a systematic approach to identifying potential risks, analyzing their possible impacts, and implementing strategies to mitigate them. In this project, risk management ensures smooth execution by minimizing disruptions and uncertainties at every stage.

5.2.1 Risk Identification

- Potential risks were identified through brainstorming sessions, review of similar past projects, and expert consultations. Key risks include data unavailability, machine learning model underperformance, difficulty in system integration, delays due to human resource constraints, and low user acceptance or satisfaction.

5.2.2 Risk Analysis

- Each identified risk was evaluated based on two factors: likelihood of occurrence and impact severity. High-probability, high-impact risks such as data quality issues and integration challenges were prioritized. A risk matrix was prepared to categorize risks into high, medium, and low levels, facilitating focused mitigation planning.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- Risk mitigation strategies include sourcing multiple nutritional databases to reduce data dependency, building modular APIs to ease integration, and maintaining a resource backup plan to cover human resource shortages. Monitoring involves periodic risk assessments, project review meetings, and maintaining a risk register. Management includes assigning ownership for each critical risk and ensuring swift escalation in case of deviations.

5.3 Project Schedule

5.3.1 Project Task Set

- The project tasks were divided into sequential and iterative modules: requirement analysis, data collection and preprocessing, machine learning model development, backend API development, frontend user interface development, integration testing, system deployment, and final user feedback analysis.

5.3.2 Task Network

- The task network defines the logical order and dependencies between tasks. Data collection must be completed before model training. Model development and backend APIs must be ready before frontend integration. Testing and deployment are dependent on the completion of both frontend and backend development. This network ensures streamlined workflow and resource optimization.

5.3.3 Timeline Chart

Week(s)	Tasks
Week 1–2	Requirements gathering, team role assignment
Week 3–4	System design finalization, technology stack selection
Week 5–6	Dataset collection and cleaning
Week 7–8	Machine Learning model selection and initial training
Week 9–10	Backend API development begins
Week 11–12	Frontend prototyping and API integration
Week 13–14	Full system integration and functionality check
Week 15–16	User Acceptance Testing (UAT) and feedback collection
Week 17–18	Bug fixes, performance optimization, UI improvements
Week 19–20	Final deployment, documentation, and submission

5.4 Team Organization

5.4.1 Team structure

- The team follows a cross-functional structure with distinct responsibilities:
 - Project Manager at the top level coordinates the activities of both development and domain experts.
 - Our team consists of four members: Anup, Aditya, Sahil, and Harsh. All members work collaboratively on every part of the

project, including report writing, implementation, frontend, backend, and other development tasks.

- A Nutrition Domain Consultant interacts regularly with the development team to ensure the scientific validity of the system outputs.

5.4.2 Management reporting and communication

- Regular progress reporting is conducted through weekly meetings, supported by online tools like Trello for task tracking and Slack for real-time communication. Formal project status reports are submitted to the academic supervisor every two weeks, summarizing completed tasks, upcoming activities, risk updates, and resource utilization.
- Our academic mentor is Dr. A. R. Deshpande Ma'am. We regularly meet her weekly or bi-weekly to seek guidance, suggestions, and feedback on our progress and implementation strategies.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

- **User Input Module:** Collects personal details such as age, weight, height, and dietary preferences using the frontend form (Streamlit).
- **Recommendation Engine Module:** Uses a Nearest Neighbors algorithm with cosine similarity to suggest food items based on the user's nutritional needs.
- **Diet Generation Module:** Aggregates recommended food items into a full-day meal plan with appropriate calorie distribution.
- **Custom Recommendation Module:** Allows users to input specific nutritional preferences (e.g., low carb, high protein) and receive tailored suggestions.
- **API Layer Module:** Built using FastAPI, it handles requests and responses between the frontend and machine learning model.
- **Data Management Module:** Loads and preprocesses the Food.com dataset for use in training and querying.
- **Visualization and Feedback Module:** Displays diet plan outputs, tracks user history, and provides options for feedback to refine future recommendations.

6.2 Tools and Technologies Used

- **Python:** Primary programming language used across the project.
- **FastAPI:** Used for building the backend API that interfaces with the recommendation engine.
- **Streamlit:** Used for building the frontend web interface.
- **Scikit-learn:** Used for implementing the Nearest Neighbors algorithm and cosine similarity.
- **Pandas and NumPy:** For data manipulation and preprocessing.
- **Matplotlib and Seaborn:** For data visualization during model evaluation and frontend display.

- **Jupyter Notebook:** For developing and testing the machine learning logic.
- **Kaggle Food.com Dataset:** Dataset used for training and generating recommendations.

6.3 Algorithm Details

6.3.1 Algorithm 1

Nearest Neighbors Algorithm (Content-Based Filtering)

This algorithm computes the similarity between food items using nutritional features (calories, carbs, fat, protein). Cosine similarity is used to find the top-N most similar recipes to the user's dietary profile. The brute-force search approach is selected due to its effectiveness on moderate-sized datasets. This method ensures recommendations are relevant to the user's input profile.

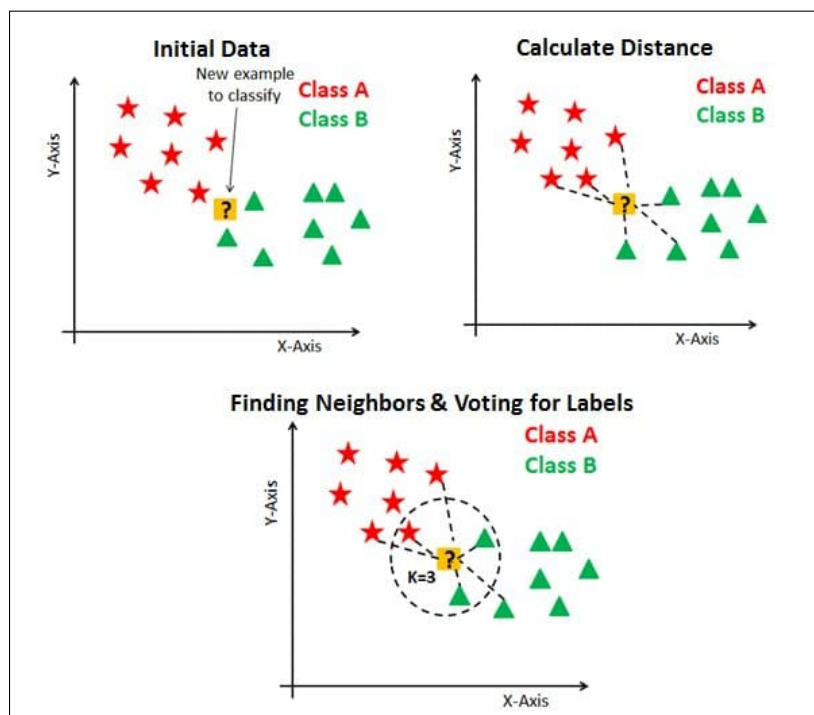


Figure 6.1: KNN Algorithm

6.3.2 Algorithm 2

Cosine Similarity Calculation

Cosine similarity is used to measure how similar two food vectors are in terms of their nutritional composition. It is defined as:

$$\cos(\theta) = \frac{A \cdot B}{||A|| \cdot ||B||}$$

Where A and B are nutritional vectors. The higher the cosine similarity value, the more similar two food items are. This helps in ranking and filtering suitable recommendations.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

- **Unit Testing:** Ensures individual components such as the recommendation function, user input validation, and API endpoints work as expected.
- **Integration Testing:** Verifies that the frontend (Streamlit), backend (FastAPI), and recommendation model work seamlessly together when deployed.
- **Functional Testing:** Confirms that the system meets all functional requirements such as user registration, meal generation, and feedback collection.
- **Performance Testing:** Checks the time taken to respond with meal recommendations under different load conditions.

7.2 Test Cases & Test Results

ID	Test Case	Input	Expected Output	Result
TC1	User Registration	Name, Email, Password (valid)	User is registered and redirected to dashboard	Pass
TC2	Diet Recommendation Based on Input	Age = 25, Weight = 70kg, Height = 170cm	Personalized meal plan generated	Pass
TC3	Custom Food Preference Entry	Preferences: High protein, Low carb	Recommendations matching user-defined nutrient levels	Pass
TC4	API Response Time	API call with valid user health data	Response within 3 seconds	Pass
TC5	Invalid Input Handling	Age = -5 (negative value)	Input validation error message displayed	Pass

Table 7.1: Compact Test Cases and Results for Diet Recommendation System

CHAPTER 8

RESULTS

8.1 Outcomes

1. The system successfully accepts user input including personal health metrics and dietary preferences.
2. Machine learning algorithms generate customized meal plans tailored to user goals such as weight loss or diabetes management.
3. The system provides nutritional breakdowns and supports updates based on user feedback and health data over time.

8.2 Screen Shots

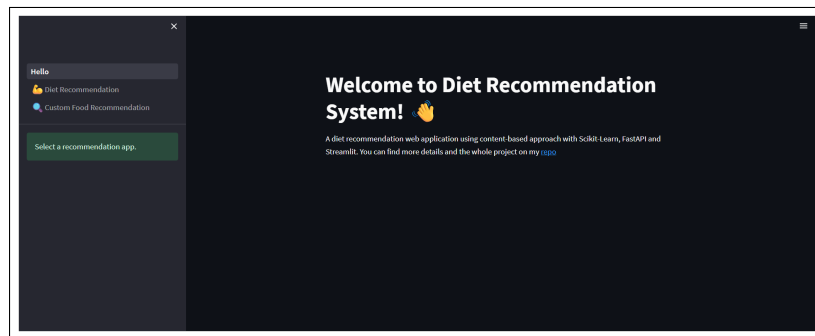


Figure 8.1: User Registration Page

This is the initial page where users register to access the diet recommendation system. The form collects basic user details such as name, email ID, and a secure password. Registration is a mandatory step to store and associate health data and dietary preferences with a specific user account. The system uses this account to track progress over time and adapt future recommendations. Input validations are implemented to ensure data integrity and prevent duplicate entries.

In this form, users provide essential health-related inputs such as age, weight, height, gender, fitness goals (e.g., weight loss, gain, or maintenance), and dietary preferences or restrictions (e.g., vegetarian, vegan, low-carb). These inputs are used to calculate nutritional requirements and generate a custom

Automatic Diet Recommendation

Modify the values and click the Generate button to see

Age: 25

Heightcm: 180

Weightkg: 75

Gender: ☒ Male ☐ Female

Activity: Little/no exercise (0-2 days/week) Moderate exercise (3-5 days/week) Extra active (very active & physical job)

Choose your weight loss plan:

Meals per day: 3

Generate

Figure 8.2: Health Information Input

meal plan. The form is user-friendly and includes error messages for invalid or missing values. The accuracy of this input significantly affects the quality of the diet recommendation output.

BMI CALCULATOR

Body Mass Index (BMI)

40.0 kg/m²

Obesity

Healthy BMI range: 18.5 kg/m² - 25 kg/m²

CALORIES CALCULATOR

The results show a number of daily calorie estimates that can be used as a guideline for how many calories to consume each day to maintain, lose, or gain weight at a chosen rate.

Maintain weight	Mild weight loss	Weight loss	Extreme weight loss
632 Calories/day	568 Calories/day	505 Calories/day	379 Calories/day
0 - 0.5 kg/week	0 - 0.25 kg/week	0 - 0.5 kg/week	0 - 1 kg/week

Generating recommendations...

Figure 8.3: Generated Meal Plan

This output screen shows the personalized meal plan generated using the Nearest Neighbors algorithm based on cosine similarity. It lists meals suitable for breakfast, lunch, and dinner along with snack suggestions. The recommendations are derived by comparing user data with items in the Food.com dataset. The plan considers user goals and restricts food items that don't match their preferences or restrictions. This interface helps users make better food choices in a structured daily format.

This screen provides a detailed breakdown of the nutrients in the recommended meal plan, including total calories, carbohydrates, fats, and proteins. It also highlights the percentage contribution of each nutrient toward

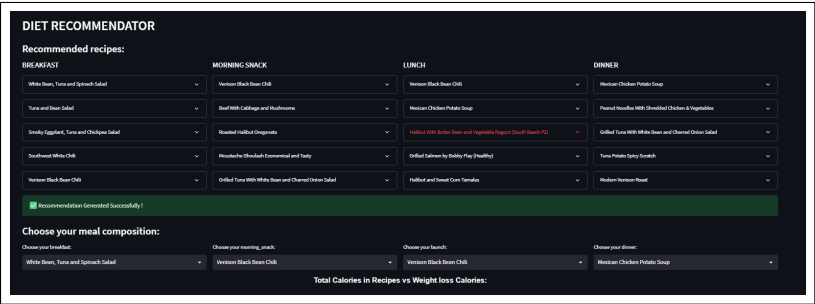


Figure 8.4: Nutritional Breakdown

the user’s daily recommended intake based on their profile. This feature is essential for users who want to monitor their diet more precisely, especially those managing medical conditions like diabetes or hypertension. It also serves as an educational tool to improve nutritional awareness.

CHAPTER 9

CONCLUSIONS

9.1 Applications

1. **Personalized Nutrition Planning:** The system can generate customized meal plans based on individual health metrics, dietary restrictions, and preferences, helping users manage conditions like obesity, diabetes, or hypertension.
2. **Health and Wellness Apps:** The recommendation engine can be integrated into health and wellness apps, providing users with daily meal suggestions that align with their fitness goals and nutritional needs.
3. **Hospital and Clinical Use:** Dietitians and healthcare providers can use the system to create specific dietary plans for patients with chronic illnesses or those recovering from surgery, ensuring they receive the right nutrients.
4. **Fitness and Weight Management Programs:** Fitness trainers and wellness coaches can integrate this system to offer diet plans that complement workout regimens and weight loss goals.
5. **Smart Devices:** The system can be integrated into wearable devices and smart kitchens, offering real-time recommendations based on the user's physical activity, health data, and available ingredients.
6. **Community and Public Health Initiatives:** Government and non-profit organizations can use this tool to promote healthy eating habits and nutritional education in communities, especially for under-served populations.

9.2 Future Work

1. **Multilingual and Cultural Adaptability:** Expanding the system to support multiple languages and tailoring recommendations based on cultural preferences would make the platform accessible to a more diverse global user base.
2. **Integration with Healthcare Systems:** Connecting the platform with healthcare providers or electronic health records could further personalize the recommendations, incorporating professional medical advice and historical health data.

3. Mobile Application Development: Developing a user-friendly mobile application could increase accessibility and convenience, allowing users to track their meals and receive recommendations on the go. These future directions will further enhance the impact and usability of the system, making it a more comprehensive tool for managing diet and health.

9.3 Conclusion

The Diet Recommendation System addresses the increasing demand for personalized dietary guidance, especially in an era where lifestyle-related diseases like obesity, diabetes, and cardiovascular conditions are prevalent. Traditional diet plans often fail to consider the unique health profiles, preferences, and goals of individuals, making it difficult for people to adhere to them long-term. By utilizing machine learning algorithms and user-specific data, this system generates personalized meal plans that are not only nutritionally balanced but also aligned with the user's health conditions, age, weight, dietary preferences, and lifestyle choices.

One of the standout features of this system is its adaptability, enabled by a continuous feedback mechanism. As users input more data or experience changes in health metrics, the system dynamically adjusts its recommendations to ensure they remain relevant and beneficial. This level of personalization makes the system a powerful tool for promoting sustainable health behavior changes.

Additionally, by incorporating real-time health data and accounting for broader socio-economic and cultural factors, the project closes a critical gap in existing diet recommendation solutions. It moves beyond generic plans and delivers actionable, tailored advice that fits diverse user backgrounds and contexts.

Ultimately, this system empowers individuals to make informed, healthier dietary choices while improving their overall well-being. In the long term, it has the potential to significantly contribute to preventive healthcare by encouraging healthier eating habits and reducing the risks associated with chronic diseases.

CHAPTER 10

APPENDIX

10.1 Appendix A

The problem statement of this project—designing a Diet Recommendation System—has been assessed for computational feasibility using standard complexity classes.

- **Problem Type:** The central computational task involves generating personalized diet plans based on constraints (health metrics, preferences, and goals), which can be mapped to a constrained optimization problem.
- **Feasibility:** Generating such a plan is similar to solving a variation of the knapsack problem (e.g., maximizing nutritional value within caloric limits), which is known to be NP-Hard.
- **Class Determination:**
 - If all user constraints are linear and the dataset size is fixed, the problem becomes polynomial-time solvable using greedy or dynamic programming approaches (i.e., P class).
 - However, with multiple conflicting constraints (e.g., dietary preferences, allergies, nutrition targets), the problem complexity increases significantly, indicating an NP-Complete nature.
- **Conclusion:** The problem lies in NP and is practically solvable using heuristic or approximation methods (e.g., Genetic Algorithms) to meet real-time response requirements.

Relevant mathematical models include constraint satisfaction problems (CSP), linear programming, and combinatorial optimization frameworks.

10.2 Appendix B

Paper Title: *Personalized Diet Recommendation Using Machine Learning*

Published In: International Journal of Creative Research Thoughts (IJCRT)

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Issue: Volume 12, Issue 3, March 2025

Link to the Paper:: <https://www.jetir.org/papers/JETIR2503579.pdf>

Authors: Aditya Bandgar, Anup Kesarwani, Harsh Lambe, Sahil Pawar

Attachments:

Certificates:



Figure 10.1: Certificate 1



Figure 10.2: Certificate 2



Figure 10.3: Certificate 3



Figure 10.4: Certificate 4



Figure 10.5: Certificate 5

10.3 Appendix C

The plagiarism report was generated using SmallSEOTools. The similarity index is summarized below:

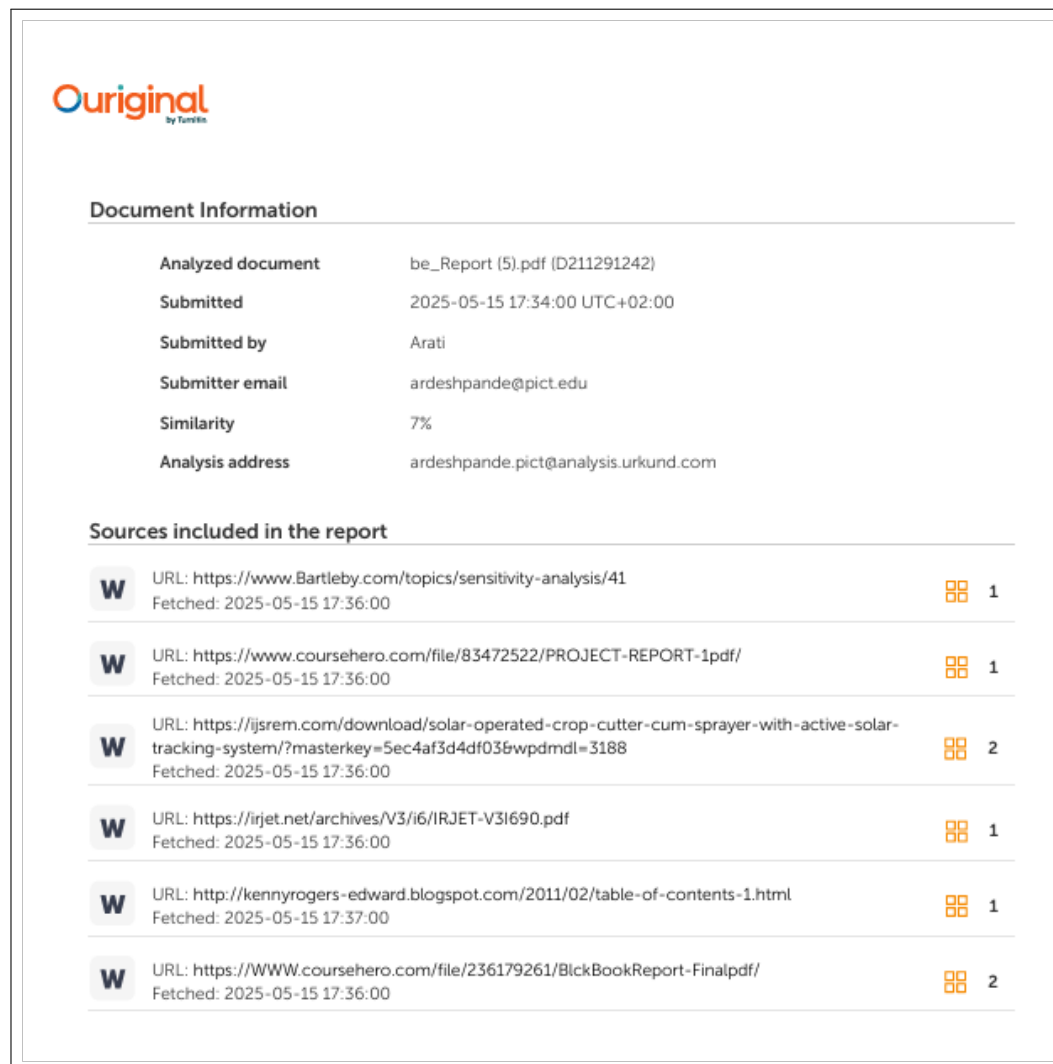


Figure 10.6: Plagiarism Report

CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
VPN	Virtual Private Network
IP	Internet Protocol
IDS	Intrusion Detection System
TCP	Transmission Control Protocol